# Decision Radar: High-Level Design

**Version:** 0.4 · **Updated:** 21 September 2026 · **Status:** Proposed Workspace-Connected MVP Design

## 1. Purpose & Core Value

Decision Radar automatically detects when new information in Google Workspace undermines prior project decisions. It continuously ingests project artifacts, monitors active assumptions against updates, and surfaces human-in-the-loop review alerts detailing what changed, affected decisions/milestones, and supporting evidence links.

## 2. Users & Core Journey

### Primary User

Project Manager (manages setup, reviews alerts, exports briefs). Team members contribute Workspace updates; viewers inspect the decision register and evidence.

### Core Journey

1. **Authorize Scope:** Admin signs in via Firebase/Google and connects a My Drive folder tree (including subfolders/supported uploads) and up to 5 recurring Meet spaces.
2. **Baseline Intake:** Decision Radar extracts initial decisions, owners, assumptions, and dependencies for human confirmation.
3. **Background Sync:** App polls Drive changes (Docs, Sheets, supported uploads) and listens to Meet transcript events via Cloud Pub/Sub.
4. **Automated Analysis:** Sync updates facts; Gemini and typed comparison rules continuously evaluate active confirmed assumptions.
5. **Review Alert:** When an assumption is challenged, an alert is queued showing side-by-side original vs. update evidence with links to native Workspace sources.
6. **Human Decision:** PM keeps, revises, or retires the decision with an explicit audit trail (no auto-rewriting of sources).
7. **Export Brief:** PM previews and exports a versioned impact brief to Google Docs.

### Demonstration Scenario (Synthetic Fixture)

- **Baseline:** Doc/Meet transcript establishes *"Launch on 20 Oct 2026, assuming vendor delivery by 10 Oct."*
- **Update:** Teammate changes vendor delivery date to 24 Oct in a Sheet; subsequent Meet transcript confirms.
- **Alert:** Delivery is delayed by 14 days, violating the assumption. Decision Radar alerts PM to review launch readiness and integration testing dependencies.

## 3. Scope & Requirements

## Key Requirements (R1–R12)

| ID | Requirement | Acceptance Behavior |
|---|---|---|
| **R1** | Workspace Connection | Connect once; recursively discover Drive folders/files & supported uploads; fetch Meet transcripts. |
| **R2** | Structured Extraction | Suggest decision, owner, rationale, typed assumptions, and source references. |
| **R3** | Human Confirmation | Draft decisions/assumptions do not enter monitoring until confirmed by an editor. |
| **R4** | Automated Change Detection | Continuously compare active assumptions against eligible updates; no-op on unchanged syncs. |
| **R5** | Evidence Traceability | Every alert includes validated original baseline and update reference links/quotes. |
| **R6** | Impact Graph | Traverse confirmed dependencies; distinguish direct challenges from downstream impacts. |
| **R7** | Review Workflow | Support Keep, Revise, Retire, Dismiss, and Request Clarification with audit history. |
| **R8** | Brief Export | Preview and export versioned, cited review briefs to Google Docs on explicit action. |
| **R9** | Idempotency & Recovery | Retries produce no duplicate decisions, alerts, or processing runs. |
| **R10** | Quality Evaluation | Benchmark detection precision, recall, citation accuracy, and review time. |
| **R11** | Connection Lifecycle | Automatic OAuth renewal, cursor recovery, token revocation handling, health metrics. |
| **R12** | Access-Controlled Evidence | Cite tabs, rows/cells, PDF pages, paragraphs, and transcript entries. Enforce viewer Workspace permissions. |

## System Limits & Out-of-Scope

### MVP Capacity Controls

- **Team Size:** Up to 10 members
- **Decisions & Assumptions:** 50 active decisions, 150 active assumptions
- **Ingestion Scope:** 250 folders, 100 connected files/transcripts, 500 update facts
- **Source Limits:** 100k chars/snapshot; uploaded files ≤ 10 MB / 50 PDF pages; tables ≤ 1,000 rows × 20 cols

### Out of Scope (MVP)

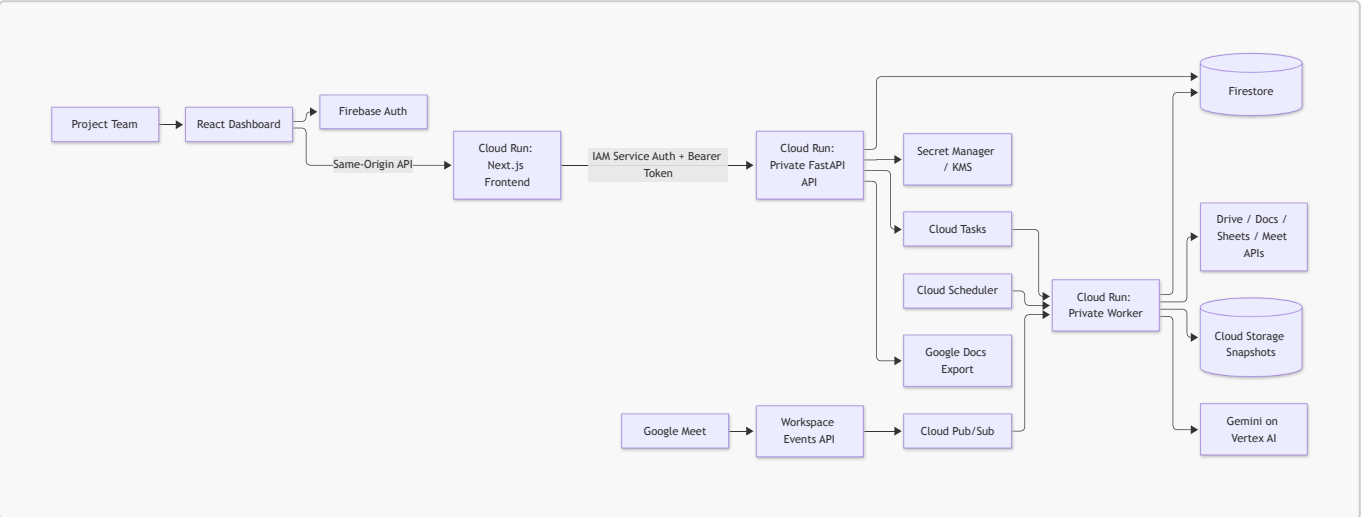
- Live call audio listening / bot joining

- Gmail / Google Chat ingestion
- Scanned/image-only PDFs requiring OCR
- Shared Drives & org-wide automatic discovery
- Auto-rescheduling or autonomous document editing

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## 4. Architecture & Component Blueprint

Decision Radar uses a single repository deployed as three Cloud Run services: a public Next.js frontend, a private Python FastAPI API, and a private Python background worker.

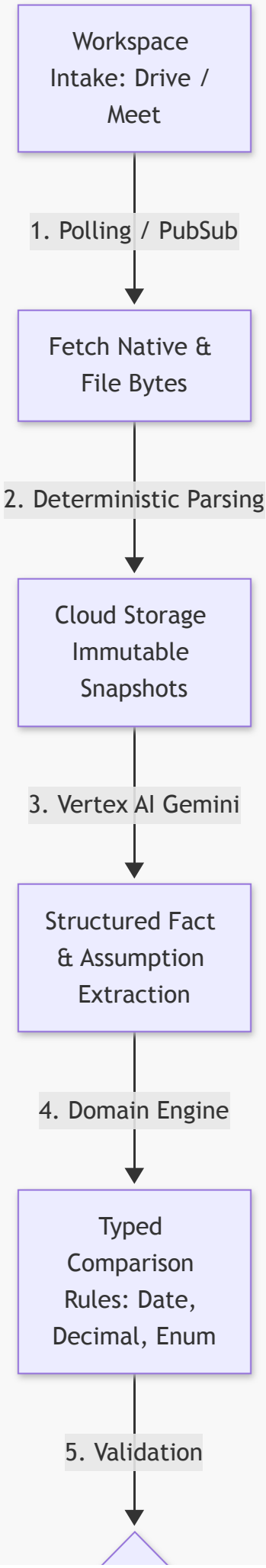


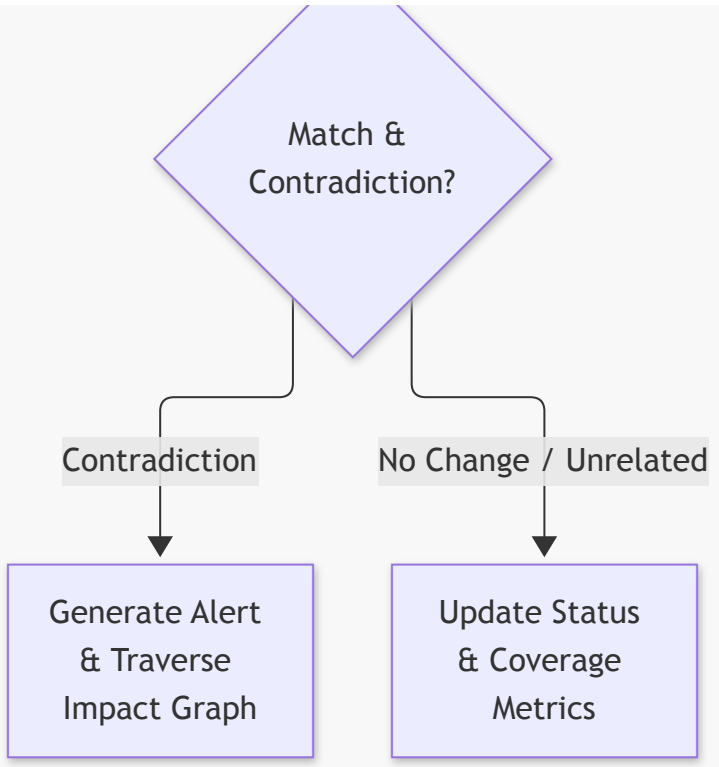

### Component Responsibilities

| Component | Stack | Primary Responsibility |
|---|---|---|
| **Dashboard** | Next.js App Router + React + TS | Setup UI, source health, decision register, review queue, brief preview. |
| **Frontend Server** | Node.js (Cloud Run) | Serve web assets & proxy same-origin `/api/v1/*` requests to API via IAM auth. |
| **API Service** | Python / FastAPI (Cloud Run) | Auth verification, authorization, validation, decision/review CRUD, Doc export. |
| **Background Worker** | Python (Cloud Run) | Drive scanning, snapshot normalization, LLM extraction, comparison, evidence verification. |
| **AI Processing** | Gemini on Vertex AI | Schema-constrained extraction and semantic comparison. |
| **State & Storage** | Firestore & Cloud Storage | App state, audit history, revision data (Firestore); immutable text snapshots (GCS). |
| **Task & Sync Engine** | Cloud Tasks & Cloud Scheduler | Stage execution, retry delivery, scheduled polling, subscription renewal. |

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## 5. Data & AI Pipeline


Workspace Intake: Drive / Meet
1. Polling / PubSub
Fetch Native & File Bytes
2. Deterministic Parsing
Cloud Storage Immutable Snapshots
3. Vertex AI Gemini
Structured Fact & Assumption Extraction
4. Domain Engine
Typed Comparison Rules: Date, Decimal, Enum
5. Validation




Ingestion Adapters

| Source | Intake Mechanism | Ingestion Behavior |
|---|---|---|
| **Google Docs** | Drive Changes API (1 min) + Docs API | Capture all tabs, structural locators (tab, paragraph), debounce active edits. |
| **Google Sheets** | Drive Changes API + 5-min Range Read | Compare typed row/cell values; detect calculated formula value changes. |
| **Uploaded Files** | Drive Changes API + Download Media | Parse text PDFs, DOCX, TXT/MD, CSV, XLSX; extract page/paragraph/row locators. |
| **Google Meet** | Workspace Events API + Pub/Sub (5 min fallback) | Process completed transcript artifacts, preserve speaker/timestamp provenance. |

## 6. Evidence, Trust & Access Control Model

- **Locators & Evidence Spans:** Immutable references citing tab IDs, structural element paths, A1 cells, PDF page numbers, DOCX paragraphs, or Meet speaker timestamps.
- **Current vs. Historical:** Changing a source updates current facts for that source ID, but preserves historical snapshots and cited quotes in audit records.
- **Coverage Statuses:** App explicitly reports status: `Needs Review`, `Evidence Unclear`, or `No Detected Issue in Assessed Sources`. Disconnected or unmapped sources surface visible coverage warnings.

## 7. Deferred Architecture Options

| Option | Status | Rationale |
| --- | --- | --- |
| **Agent Search / Vertex AI Search** | Optional MVP Extension | Exhaustive assumption comparison is simpler and avoids retrieval misses at small scale. |
| **ADK / Agent Engine** | Deferred | Pipeline follows known deterministic stages with explicit human checkpoints. |
| **Document AI** | Deferred | Standard text parsing covers initial file formats; scanned OCR PDFs are out of scope for MVP. |

## 8. Security & Compliance Checklist

- ☑ **Strict Authentication:** Firebase ID tokens verified on every API request.
- ☑ **Per-User Source Access Verification:** Server verifies viewer's native Workspace read permissions before displaying derived content, alerts, or briefs.
- ☑ **Isolated Service Identities:** Next.js frontend has Cloud Run Invoker role for API only. Browser never touches database, storage, or AI credentials.
- ☑ **Least-Privilege Scopes:** Backend uses `drive.readonly` and `meetings.space.readonly` for intake. Incremental `drive.file` requested only during user-initiated Doc export.
- ☑ **Token Security:** OAuth client secrets stored in Secret Manager; refresh tokens encrypted with Cloud KMS.

## 9. Quality & Performance Targets

| Metric | Target | Measurement Method |
| --- | --- | --- |
| **Alert Precision** | >= 90 | Correct direct alerts on held-out synthetic test set (30 scenarios). |
| **Alert Recall** | >= 85 | Labeled challenged assumptions producing an alert. |
| **Citation Integrity** | 100% | Published references resolving to exact stored quote spans. |
| **Assessment Latency** | < 60s (p95) | Processing latency for <= 10 decisions & <= 20k chars. |
| **Ingestion Freshness** | < 3min Docs/Sheets, < 2min Meet | Delay from native source edit/transcript event to capture. |

## 10. Delivery Plan & Risk Matrix

### 3-Week Implementation Timeline

| Timeline | Deliverables | Exit Criteria |
|---|---|---|
| **Days 1–2** | Prerequisites, Next.js/API scaffold, OAuth offline grants. | Read Docs/Sheets/Meet native APIs server-to-server. |
| **Days 3–5** | Recursive folder bootstrap, Drive cursor, Docs & upload snapshots. | Nested file edit/upload triggers cited baseline analysis. |
| **Days 6–7** | Mapped Sheets/CSV/XLSX tables & typed comparison rules. | Cell date/budget change surfaces challenged assumption. |
| **Days 8–10** | Meet Pub/Sub event ingestion & periodic reconciliation. | Transcript arrival automatically fetches and extracts facts. |
| **Days 11–14** | Review workflow, impact graph traversal, Docs export. | End-to-end review brief preview and export verified. |
| **Days 15–17** | Evaluation suite runs & usability fixes. | Held-out benchmark quality gates passed. |
| **Days 18–21** | Polish, operational hardening & rehearsal. | Repeatable demo ready with documented limits. |

## Top Risks & Mitigations

| Risk | Mitigation |
|---|---|
| **OAuth Test-Mode Expiry (7-day limits)** | Re-authorize grants before demo; implement explicit token renewal checks. |
| **False Alerts & Hallucinations** | Require exact evidence matching + typed deterministic date/decimal rules. |
| **Sheet Schema Drift** | Pause affected table and surface a coverage gap when header/columns change. |